

Teaching Monocular Depth the Metre with the Phone’s Own LiDAR: Fine-Tuned vs. Pretrained Depth Anything V2 for Room Reconstruction, Measured Against a Laser Scanner

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Abstract—Most phones have no LiDAR and must scan rooms with monocular depth, yet off-the-shelf “metric” models such as Depth Anything V2 (DA-v2) Metric-Indoor [1], trained on synthetic images, misplace a real room by 54.3 cm. We ask whether fine-tuning such a model with the phone’s own LiDAR depth, with no laser scanner involved, makes it more accurate than its pretrained self, measured against the Faro Focus S70 laser scanner [2] that provides the ground truth of ARKitScenes [3]. We fine-tune only the DPT head [4] (the depth decoder; the image encoder stays frozen) on 24 rooms with iPad LiDAR labels, giving FT-LiDAR-24, and compare it with the pretrained model in one pipeline—per-frame depth fused into a truncated signed distance function (TSDF) volume [5], then meshed—where the depth source is the only variable, on rooms never used for training: six with a laser-scanned reference mesh and 20 more evaluated per frame. FT-LiDAR-24 lowers the Chamfer distance [6] (mean nearest-point distance between the reconstructed and reference meshes, both ways) from 54.3 cm to 15.7 cm ($3.5\times$, better in every room) and AbsRel [7] (mean relative depth error) from 0.410 to 0.130 (better in 19/20 rooms), with no calibration or sensor at run time; a laser teacher is not measurably better. Its remaining error is a scale that swings from view to view, which fine-tuning does not remove, so it still trails iPad LiDAR (3.0 cm) and falls short of overview accuracy ($\geq 90\%$ of the room within 10 cm).

Index Terms—monocular depth estimation, transfer learning, metric depth, LiDAR supervision, TSDF fusion, 3D room reconstruction, ARKitScenes

I. INTRODUCTION

Scanning a room into a 3D model is already a consumer feature on iPhone Pro and iPad Pro (e.g., Apple RoomPlan [8]), and it relies on the *LiDAR sensor* for depth in metres. Devices with LiDAR are, however, a minority: non-Pro iPhones and almost every Android phone have none. Their alternative is *monocular depth estimation*—depth predicted from a single colour image—which has advanced rapidly from MiDaS [9] to Depth Anything [1], [10]. A product team must therefore know **by how many centimetres the room is wrong if LiDAR is replaced by a monocular model, and which use cases that still serves**. Standard per-image benchmarks [7], which average the depth error over the pixels of single images, do not answer this, and end-to-end reconstruction systems [11]–

[13] cannot, because they design the whole pipeline around their own model.

Some recent models output metres directly [1], [14], [15], but they are trained on images that do not come from a phone camera—DA-v2 Metric-Indoor on rendered rooms of the Hypersim dataset [16]—and in our experiments a room reconstructed with DA-v2 Metric-Indoor is 54.3 cm wrong on average. The direct remedy is to *fine-tune* the model on depth from the target device, which raises the question of where these labels come from: a survey-grade laser scanner [2], which few app makers own, or the *LiDAR already inside Pro devices*, whose depth an app can collect during normal use. If the latter works, devices with LiDAR can teach those without it.

We therefore ask whether DA-v2 Metric-Indoor fine-tuned with iPad LiDAR depth, with no laser scanner at training time, is more accurate than the same model pretrained, when both are measured against the Faro laser scanner that provides the ground truth of ARKitScenes [3] on rooms never used for training. The study has four objectives: (1) to fine-tune the model on iPad LiDAR depth from 24 rooms, giving FT-LiDAR-24, and compare it with the pretrained model on held-out rooms, counting it as better if its mean error is lower and the per-room difference is significant at the 5% level under an exact Wilcoxon signed-rank test [17]; (2) to measure whether the source of the labels matters, by comparing Faro with LiDAR labels on the same 10 rooms (FT-Faro-10 and FT-LiDAR-10), 10 with 24 rooms of LiDAR labels, and a model that uses the camera’s focal length instead of in-domain labels (Depth Pro [15]); (3) to place FT-LiDAR-24 among the depth sources available at run time, namely iPad LiDAR and a relative model scaled per frame on sparse points or once per room; and (4) to find the error that fine-tuning leaves, by fitting a depth scale to every frame and testing whether it follows time, which would indicate drift, or what the view shows.

Every comparison is scored in two ways. As a room mesh, it is scored by the Chamfer distance [6], the mean distance

from each surface to the nearest point of the other, and by the F-score at 5 cm (F@5 cm) [18], which combines the share of the reconstruction lying within 5 cm of the reference with the share of the reference reconstructed within 5 cm. Per frame, it is scored by AbsRel [7], the mean relative depth error, and by the share of pixels whose predicted depth is within a factor of 1.25 of the true depth, which we call within-1.25 $\times$ [7]. For objective (3), the share of the reference reconstructed within 2, 5 and 10 cm stands for the accuracy needed for renovation measurement, furniture placement and a room overview, respectively, and a use case counts as served when that share reaches 90 %.

The scope is as follows. The data are home rooms of ARKitScenes captured with one iPad Pro model (colour images of 640×480 pixels, LiDAR depth of 256×192); the ground truth is the Faro depth distributed with the dataset, and iPad LiDAR is never used as ground truth. Only the DPT head [4], the depth decoder of DA-v2 Metric-Indoor Large, is fine-tuned, with one training recipe, while its image encoder stays frozen. Six test rooms are evaluated as meshes and per frame, and 20 further rooms per frame only, all disjoint from the rooms used for training and checkpoint selection. Every depth source passes through the same offline pipeline—per-frame depth, scale alignment, fusion into a truncated signed distance function (TSDF) volume [5], meshing—so the depth source is the only variable, and because every source and the reference share the camera poses recorded by ARKit, pose error is excluded. Learned reconstruction systems [11]–[13], other phone cameras including Android, real-time inference on the device and pose estimation are outside the scope.

Each objective has a matching expected benefit. The first tells an app maker whether data it can already collect from Pro-device users improves room scans on devices without LiDAR, with no calibration or extra sensor at run time; the second, whether labelling needs a laser scanner at all; the third, which product features a device without LiDAR can offer, together with a pipeline in which a new depth source is evaluated by changing one configuration entry; and the fourth, where to invest next—per-frame scale, training data or model size.

II. BACKGROUND AND RELATED WORK

This section explains the theory the study rests on—why one image does not fix depth in metres, how the model we fine-tune is built and trained, and how depth maps become a room—and places the study among prior work on each point.

A. Depth from a single image

In the pinhole model of a camera [19], the size of an object in the image is proportional to its real size divided by its distance, multiplied by the focal length of the lens. A room twice as large seen from twice as far therefore gives the same image, so geometry alone cannot recover depth in metres from one image; a network can only infer it from learned cues, such as the usual height of a door or a table, and even these cues depend on the focal length, because the same object at the same distance looks larger through a longer lens. Eigen et

al. [7] trained an early deep network for this task and, because of this ambiguity, measured its error on the logarithm of depth in a way that separates an error in the overall scale of the scene from an error in its shape. A weighted form of this scale-invariant logarithmic loss, known as SiLog, also trains ZoeDepth [14] and the fine-tunes of this study.

Two families of models handle the missing scale differently. Relative models give it up: MiDaS [9] trained one network on a mixture of datasets whose depths came from different sensors and units, which is possible only if the loss ignores the scale and offset of each image. Its output is therefore inverse depth (disparity) that is correct only up to an unknown scale and offset per image, and these two numbers must be found from other data before the output gives metres. Depth Anything [10] and Depth Anything V2 [1] (DA-v2) keep this output but train on far more images. Metric models output metres directly by learning the scale from labelled data. ZoeDepth [14] attaches metric heads, fine-tuned on NYU Depth v2 [20] and KITTI [21], to a relative model, and DA-v2 releases metric variants fine-tuned from its relative model, the indoor one (DA-v2 Metric-Indoor) on the rendered rooms of Hypersim [16]. A further line addresses the focal length: Metric3D [22] maps every image to a canonical camera with a fixed focal length, UniDepth [23] predicts the camera together with the depth, and Depth Pro [15] estimates the focal length when it is not given. These works report per-image metrics on benchmarks such as NYU Depth v2 and KITTI. Because such metrics average over independent images, they cannot show whether a model’s scale stays the same across the frames of one scan, which is what matters once the frames are fused into one room.

B. Architecture and training of Depth Anything V2

DA-v2 consists of an image encoder and a depth decoder. The encoder is DINOv2 [24], a vision transformer (ViT) [25]: it cuts the image into small square patches and lets every patch draw information from every other patch through a mechanism called self-attention. DINOv2 was pretrained without labels on a large curated collection of images, so that its features serve many tasks. The decoder is the Dense Prediction Transformer (DPT) [4], which takes features from several encoder layers, reassembles them into image-like maps at several resolutions and fuses these into one dense depth map. The released code splits the decoder into a neck, which reassembles and fuses, and a head, which outputs depth; as in the Introduction, “DPT head” in this paper means both.

DA-v2 is itself trained with a teacher. A large model is first trained on synthetic images only, whose depth is exact; it then labels about 62 million real images that have no depth, and the released models learn from these model-made labels, called pseudo-labels [1]. The DINOv2 authors report that depth can be predicted well with the encoder kept frozen and only a DPT decoder trained on top [24]. Adapting a model trained on one task or domain to another in this way is called transfer learning [26]; training only the decoder, as we do, needs much

less memory and keeps the general visual features that the encoder learned from millions of images.

C. Training on sensor depth

Supervised depth models have long been trained on depth measured by a sensor rather than drawn by people: NYU Depth v2 provides indoor depth from a Microsoft Kinect camera [20], KITTI provides street depth from a laser scanner mounted on a car [21], and Eigen et al. [7] trained on both. What differs between such studies is the quality of the sensor. On the iPhone Pro and iPad Pro, ARKit combines the LiDAR measurements with the colour image into a depth map of 256×192 pixels with a confidence value for every pixel [3]. Studies that tested this LiDAR against established surveying methods found it accurate to about 1 cm on objects larger than 10 cm [27] and adequate for rapid architectural mapping at a scale of 1:200 [28], which makes it a plausible teacher. ARKitScenes [3] provides this depth together with the depth of a Faro survey laser scanner [2] for the same rooms, registered to the same camera frames, and uses the Faro depth as ground truth for a task in which the LiDAR depth is upsampled with the help of the colour image. Prompt Depth Anything [29], the closest work to ours in data, likewise feeds iPhone LiDAR depth into a Depth Anything model as an extra input. In both, the LiDAR must be present whenever the model runs; in this study it is used only at training time, so the deployed model needs nothing but a colour image. Whether this noisier teacher helps a large pretrained model as much as a laser scanner does, when both are judged against an independent laser reference, is the subject of objectives (1) and (2).

D. From depth maps to a room

Fusing frames requires the camera pose of each frame, that is, its position and orientation. ARKit tracks the pose by visual-inertial odometry (VIO), which follows image features from frame to frame and combines them with the device’s motion sensor [30]. Because the motion sensor measures acceleration in metric units, the tracked feature points have positions in metres, and such sparse points can supply, frame by frame, the scale and offset that a relative model lacks. Combining learned depth with the sparse geometry of a tracking system goes back to CNN-SLAM [31], and MonoSDF [32] fits a scale and an offset to every image before using monocular depth to guide surface reconstruction; the sparse-point alignment of Section III follows this line.

The room itself is built by fusion into a truncated signed distance function (TSDF) [5]. Space is divided into small cubes called voxels; each voxel stores its distance to the nearest observed surface, positive in front of the surface and negative behind it, cut off (truncated) beyond a few voxels, and this value is averaged over every frame that sees the voxel. The surface is then extracted where the average crosses zero, with the marching cubes algorithm [33]. KinectFusion [34] made this fusion run in real time on Kinect depth, and we use the implementation in Open3D [35]. Averaging suppresses noise that is random from frame to frame, but frames that disagree in

scale place the same wall at different distances, and the average then blurs or thickens it; the consistency of scale across frames, which per-image metrics do not measure, therefore shapes the room. TSDF fusion also does not depend on where the depth in metres comes from, which makes it a fair fixed stage for every depth source.

Learned reconstruction systems replace parts of this pipeline with networks. Atlas [11] predicts the TSDF directly from features of many posed images, NeuralRecon [12] does so incrementally and in real time, and SimpleRecon [13] estimates depth from several neighbouring frames before classic fusion; all are evaluated on ScanNet [36], a large collection of indoor colour-and-depth scans. Each needs several frames per prediction and a network trained for its own pipeline, so they answer a different question from ours—what room a single-image model yields when nothing else in the pipeline changes—and we do not compare against them.

E. Measuring a reconstructed room

A reconstructed room is compared with a reference surface through distances between points sampled on both. The Chamfer distance [6] averages, in both directions, the distance from each point to the nearest point of the other surface. Tanks and Temples [18] instead fixes a distance threshold and reports precision, the share of the reconstruction within the threshold of the reference; recall, the share of the reference within the threshold of the reconstruction; and their harmonic mean, the F-score. A threshold ties the score to a use, which is how the Introduction links 2, 5 and 10 cm to renovation measurement, furniture placement and a room overview. Per frame, we use AbsRel and within-1.25 $\times$ from the single-image literature [7].

F. Position of this study

In summary, prior work reports the per-image accuracy of depth models on standard benchmarks, adapts metric models with labels from synthetic images or from the sensors of public datasets, feeds mobile LiDAR to a model at run time, or tests mobile LiDAR itself against surveying methods. We did not find a study that combines these threads as this one does: it uses the LiDAR of a consumer device only as a training teacher, compares it with a survey laser scanner as teacher on the same rooms, and judges both against an independent laser reference, in centimetres on fused rooms as well as per frame.

III. SYSTEM DESIGN

The research question (“change only the depth source—how much changes?”) leads to one design rule: every depth source passes through the same code after the point where it is produced, and whatever can be swapped is swapped from a YAML config, so one result table is one config file (Fig. 1).

DepthSource. An interface `get_depth(frame)` returning an (H, W) depth map and a flag `is_metric`. Implementations read Faro depth (`gt`), ARKit LiDAR (`lidar`) or run a monocular network (`mono`: DA-v2, MiDaS, fine-tuned checkpoints) on the upright RGB frame.

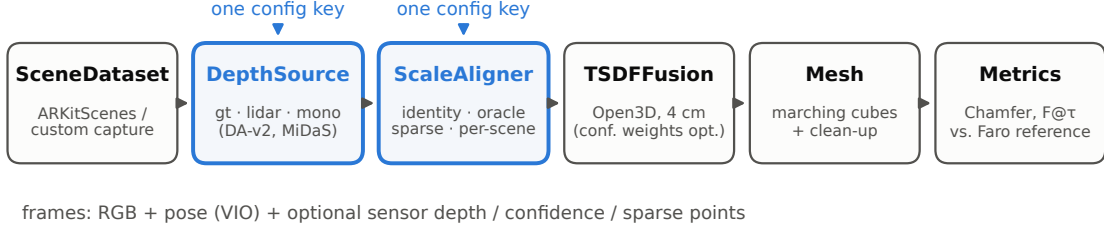


Fig. 1. Pipeline. The two highlighted stages are the only things that change between rows of a result table; everything to their right is fixed.

TABLE I

SCALE ALIGNERS. EVERY NON-IDENTITY ALIGNER FITS (s, t) IN INVERSE DEPTH WITH TRIMMED LEAST SQUARES.

Aligner	Fitted on / meaning
identity	nothing; for Faro, LiDAR and metric models
oracle_frame	that frame’s GT; the model’s ceiling (not deployable)
per_scene	GT of 10 frames spread over the scan, fitted once; upper bound of a one-off calibration
sparse_points	200 LiDAR pixels per frame at confidence 2, a proxy for VIO feature points; upper bound of a VIO-based system

ScaleAligner. A relative model is correct only up to an affine map, and *how* that map is obtained is a system decision measured separately from the model (Table I). Affine-invariant models are affine in *disparity*, so every aligner fits $1/gt \approx s \cdot (1/pred) + t$ by trimmed least squares and inverts back; on the development scene this change alone moved mono+oracle from 15.2 cm to 5.4 cm. Metric sources must use *identity*; the pipeline refuses to build otherwise.

Fixed stages. Open3D ScalableTSDFVolume with 4 cm voxels, truncation $3 \times$ voxel and depth range 0.1–5 m; every source is resampled to the LiDAR resolution (256×192) for fairness, and only pixels where Faro has depth are integrated, so no source is penalised for seeing beyond the reference. Marching cubes and small-cluster removal follow. Poses are the dataset’s ARKit VIO trajectory for every row.

IV. EXPERIMENTAL SETUP

A. Dataset and reference

ARKitScenes [3] is the only dataset that provides, for the same room, RGB video from a real device (iPad Pro, 640×480), that device’s LiDAR depth (256×192) and Faro laser depth registered to the camera poses (1920×1440 , ~ 4 fps). **Ground truth is the Faro data as distributed.** In 2D we compare per-frame depth with Faro depth directly (nearest-neighbour down-sampling, so every GT value is a measured one). In 3D, since no laser mesh is shipped, we fuse Faro depth from every frame at 1 cm voxels once per scene into a reference mesh. The *gt* row at 4 cm therefore has non-zero error—the pipeline’s own discretisation loss—and because the reference uses the same poses as every row,

TABLE II

TEST SCENES (3D AND 2D). FRAMES = FARO FRAMES EVERY ROW IS EVALUATED ON.

video_id	Room	Extent (m)	Frames
42444474	kitchen/dining, glass (dev. scene)	$9.9 \times 5.1 \times 4.2$	236
47333774	standard bedroom	$5.1 \times 4.9 \times 3.3$	375
47115299	large living room	$8.1 \times 6.1 \times 2.5$	324
42897743	bathroom (mirror, tiles)	$8.0 \times 3.4 \times 3.6$	448
47429736	bedroom + closets	$6.6 \times 6.4 \times 2.7$	504
45261556	kitchen (glossy cabinets)	$7.1 \times 6.1 \times 2.4$	465

pose drift is excluded and all differences are due to the depth source. iPad LiDAR is never used as ground truth.

The six test scenes (Table II) span standard, large and glossy/mirror rooms. Scene 42444474 was used during development, so its numbers are not held-out; per-scene results show the conclusions hold on the other five.

B. Fine-tuning protocol

Scenes disjoint from the test set at the *visit_id* level were drawn with seed 0: 10 training scenes with both Faro and LiDAR, 14 more with LiDAR only, and 3 Validation-fold scenes for checkpoint selection. Three fine-tunes share one recipe: **FT-Faro-10** uses Faro labels on the 10 scenes (3594 frames), **FT-LiDAR-10** iPad LiDAR (confidence ≥ 1) on the same 10 scenes (8656 frames), and **FT-LiDAR-24, the main model**, LiDAR on all 24 scenes (19 896 frames). FT-Faro-10 and FT-LiDAR-10 do not share frames: Faro exists at ~ 4 fps, LiDAR with every colour frame (we take every third, ~ 10 fps), so FT-LiDAR-10 sees $\sim 2.4 \times$ more distinct views than FT-Faro-10. The DINOv2 encoder is frozen; only the DPT neck and metric head train, with SiLog loss, AdamW ($1r \times 10^{-5}$, cosine schedule), batch 8 (2×4 accumulation) on 518×518 crops, for 6000 steps (FT-Faro-10, FT-LiDAR-10) or 12 000 (FT-LiDAR-24), keeping the checkpoint with the lowest validation AbsRel. A 6000-step run takes about an hour on an RTX 3070. At test time the fine-tuned models are plain DepthSource rows: RGB in, metres out, *identity* aligner, same input preparation as the pretrained model.

C. Metrics and experiments

3D: 200k points sampled on both meshes; accuracy = mean $pred \rightarrow ref$ distance, completeness = $ref \rightarrow pred$, Chamfer =

TABLE III
EXPERIMENTS. EXP. 6–7 ANSWER THE MAIN QUESTION; EXP. 1–5 GIVE CONTEXT.

Exp.	Variable (everything else fixed)
6	pretrained vs. FT-LiDAR-24 ; ablation FT-Faro-10 / FT-LiDAR-10 / Depth Pro (6 scenes, 3D + 2D)
7	pretrained / FT-LiDAR-24 / LiDAR on 20 further rooms (2D)
1	depth source $\times$ aligner: gt / lidar / DA-v2 L + {oracle, sparse, per_scene} / metric
2–5	voxel size; frame stride; model size; LiDAR confidence

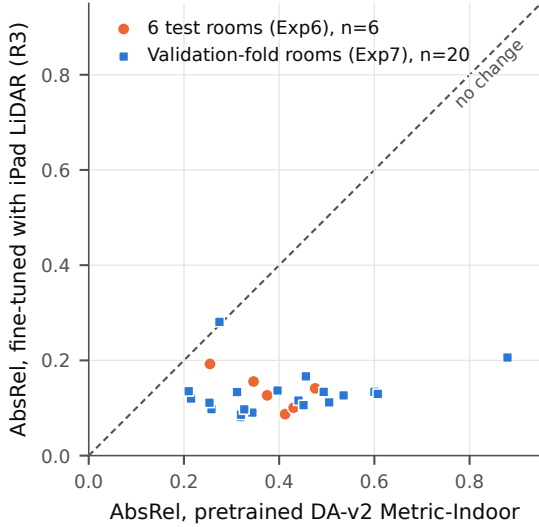


Fig. 2. Per-room AbsRel against Faro: pretrained (x) vs. FT-LiDAR-24 (y), six test rooms and 20 Validation-fold rooms. Below the diagonal = the fine-tune is better.

their mean; precision/recall/F-score at $\tau = 2/5/10$ cm, with F@5cm as headline (5 cm $\approx$ furniture-placement tolerance). **2D**: per-frame AbsRel, within-1.25 $\times$ and RMSE against Faro depth; pixels are selected by the ground truth (0.1–5 m) and predictions clipped to that range, averaged per room then over rooms. Table III lists the experiments; numbers are mean $\pm$ std over the six scenes. **Exp. 7** adds 20 Validation-fold rooms (seed 0, one video per visit, excluding every visit of the training, validation and test sets) evaluated in 2D only on every second Faro frame; these rooms decided nothing.

V. RESULTS

All numbers are generated from the per-room metrics.json files; units are cm and F is F@5cm.

A. Main result: FT-LiDAR-24 vs. the pretrained model

The two rows differ only in the weights of the DPT neck and metric head; at test time both see the same frames and input with no alignment of any kind, and no room here was used for training or checkpoint selection.

- 1) **FT-LiDAR-24 beats the pretrained model in every test room.** Chamfer 54.3 cm $\rightarrow$ 15.7 cm (3.5 $\times$; 38.6 $\pm$ 8.6 cm better per room, 6/6 rooms, exact

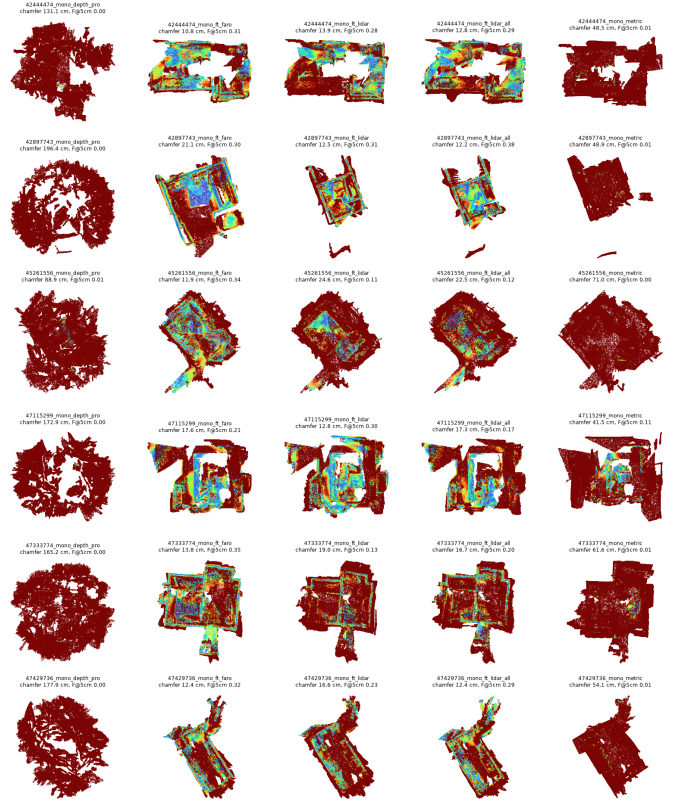


Fig. 3. Exp. 6 top-down error maps (rows = scenes; columns = Depth Pro, the three fine-tunes and the pretrained metric model, as labelled above each panel; colour = distance to the reference, 0–10 cm). The pretrained column is red almost everywhere—the room is built at the wrong distance; after fine-tuning the error concentrates at object edges and far walls.

Wilcoxon $p = 0.031$, the smallest attainable at $n = 6$) and F@5cm 0.03 $\rightarrow$ 0.24 (Tables IV, V). Accuracy falls more than completeness: the pretrained model builds the room too far away (15–31 % over-estimation, Section VI-B); FT-LiDAR-24 brings surfaces back near where they belong.

- 2) **The same holds on rooms that took part in nothing.** On the 20 Validation-fold rooms AbsRel drops 0.410 $\rightarrow$ 0.130 (3.2 $\times$; better in 19/20 rooms, exact Wilcoxon $p = 3.8 \times 10^{-6}$) and the within-1.25 $\times$ share rises 0.333 $\rightarrow$ 0.827 (Table VI, Fig. 2). The one exception, 48018730 (AbsRel 0.275 $\rightarrow$ 0.281), is one of the pretrained model's best rooms; in the other 19, FT-LiDAR-24's AbsRel is 21–64 % of the pretrained one.
- 3) **It does not reach a sensor.** FT-LiDAR-24 is $\sim$ 13 cm behind iPad LiDAR in Chamfer and 9.7 cm behind a per-frame fit on 200 sparse points; fine-tuning fixes the scale of the *domain*, not of the *frame* being looked at (Section VI-A).

B. Teacher ablation and a no-fine-tuning baseline

The laser teacher is not measurably better than the LiDAR teacher. FT-Faro-10 14.6 cm, FT-LiDAR-24 15.7 cm

TABLE IV

MAIN RESULT ON THE SIX TEST ROOMS: RGB ONLY, NO ALIGNMENT, MEASURED AGAINST FARO. THE TWO REFERENCE ROWS (EXP. 1) NEED A SENSOR AT RUN TIME.

Model	Teacher	Chamfer	Acc.	Compl.	F@5	R@10	AbsRel	within-1.25×	RMSE
DA-v2 Metric-Indoor, pretrained	— (Hypersim)	54.3 ± 10.6	70.3	38.3	0.03	0.10	0.382	0.277	49.3
+ fine-tune FT-LiDAR-24	iPad LiDAR, 24 rooms	15.7 ± 4.0	17.3	14.0	0.24	0.51	0.134	0.812	20.2
ref.: iPad LiDAR (sensor)	—	3.0 ± 0.6	2.2	3.9	0.93	0.97	0.018	0.997	3.9
ref.: DA-v2 L + 200 sparse pts/frame	—	6.0 ± 1.7	6.4	5.5	0.74	0.91	0.036	0.985	7.4

TABLE V

EXP. 6 PER TEST ROOM (ORDER OF TABLE II): CHAMFER (CM) AND ABSREL.

	4244	4733	4711	4289	4742	4526	mean
Depth Pro, Chamfer	131.1	165.2	172.9	196.4	177.9	88.9	155.4
pretrained, Chamfer	48.5	61.6	41.5	48.9	54.1	71.0	54.3
FT-Faro-10, Chamfer	10.8	13.8	17.6	21.1	12.4	11.9	14.6
FT-LiDAR-10, Chamfer	13.9	19.0	12.8	12.5	16.6	24.6	16.6
FT-LiDAR-24 , Chamfer	12.8	16.7	17.3	12.2	12.4	22.5	15.7
pretrained, AbsRel	0.375	0.346	0.255	0.412	0.476	0.430	0.382
FT-LiDAR-24 , AbsRel	0.127	0.156	0.193	0.087	0.141	0.101	0.134

TABLE VI

EXP. 7: 2D ONLY, 20 VALIDATION-FOLD ROOMS NEVER USED FOR ANYTHING ELSE (2,090 FRAMES); MEAN ± STD OVER ROOMS.

Row	AbsRel	within-1.25×	RMSE (cm)
DA-v2 Metric-Indoor, pretrained	0.410 ± 0.164	0.333 ± 0.216	50.9
+ fine-tune FT-LiDAR-24	0.130 ± 0.046	0.827 ± 0.135	18.4
ref.: iPad LiDAR (sensor)	0.019 ± 0.002	0.997 ± 0.002	3.8

and FT-LiDAR-10 16.6 cm all lie within the across-room std (4.0–4.7 cm), and no row wins every room (FT-Faro-10 beats FT-LiDAR-24 in three rooms, FT-LiDAR-24 wins two, one tie; Table V, Fig. 3). Because FT-Faro-10 and FT-LiDAR-10 also differ in the number of views, the honest claim is that a phone’s depth sensor can replace the laser as the teacher at no measurable cost on these rooms—not that the two are equal. Fourteen extra LiDAR-only rooms (FT-LiDAR-24 vs. FT-LiDAR-10) do not help measurably either; we keep FT-LiDAR-24 as the main model because it was best on validation and is the only recipe that scales without a laser.

Knowing the camera does not rescue metric depth. Depth Pro [15], given the true focal length and no fine-tuning, reaches 155.4 cm (F@5 cm 0.00), worse than the pretrained DA-v2 Metric row: it predicts a 48° field of view where the camera has 62°, and even after removing all scale per frame its AbsRel is 0.24–0.54. We verified the wrapper on a synthetic room with exact ground truth; the claim is about this model on this low-resolution indoor input, not about intrinsics-conditioned models in general.

C. Context: the price of each depth source (Exp. 1)

- 1) **The pipeline loses ~2 cm; LiDAR adds ~1 cm** (0.98 ± 0.25 cm, same in every room). The two differ clearly only at $\tau = 2$ cm: iPad LiDAR is wrong at the 2–4 cm level.

TABLE VII

EXP. 1: SAME PIPELINE, ONLY DEPTH.SOURCE / DEPTH.ALIGNER CHANGE (6 SCENES).

Source	Aligner	Chamfer	F@2	F@5	AbsRel
Faro (gt)	identity	2.1 ± 0.5	0.86	0.97	0
ARKit LiDAR	identity	3.0 ± 0.6	0.54	0.93	0.018
DA-v2 L	oracle_frame	5.3 ± 1.3	0.46	0.79	0.032
DA-v2 L	sparse_points	6.0 ± 1.7	0.38	0.74	0.036
DA-v2 L	per_scene	22.9 ± 2.1	0.08	0.23	0.237
DA-v2 Metric	identity	54.3 ± 10.6	0.01	0.03	0.382

TABLE VIII

SECONDARY EXPERIMENTS (CHAMFER IN CM, 6 SCENES).

Exp.	Result
2 voxel (gt)	2 / 4 / 8 cm: 1.2 / 2.1 / 2.8; 4 cm is used throughout (fusion 1–3 s per scan at every size)
3 frame stride	GT accuracy is constant (1.2–1.3 cm); fewer frames only lose coverage: F@5 cm 0.97 at 4 fps, 0.91 at 0.8 fps, 0.83 at 0.4 fps
4 model size (oracle)	DA-v2 L / S / MiDaS small: 5.33 / 6.35 / 13.24 at 871 / 121 / 27 ms per frame
5 LiDAR confidence	masking or weighting changes Chamfer by ≤ 0.03 cm; keeping only confidence 2 loses completeness

- 2) **Monocular depth with the right scale per frame (oracle) is 2.29 ± 1.11 cm behind LiDAR**, and **200 sparse metric points per frame recover it** (0.6 cm from the oracle, $3.8\times$ better than one-off calibration).
- 3) **One calibration per scene is $4.5\times$ worse than the oracle, uniformly** (19.9–26.0 cm in every room): a property of the model family, not of a failing room (Section VI-B).
- 4) **The pretrained metric model is worst** (54.3 cm): the surfaces are all there, in the wrong place. Kitchen 45261556 is among its best shapes under the oracle (4.5) and its worst metric room (71.0): right shape, wrong scale. This is the baseline FT-LiDAR-24 improves on.

D. Secondary findings (Exp. 2–5)

Table VIII summarises the rest. A $13\times$ smaller model (DA-v2 Small) costs only 1.0 cm for a $6\times$ speed-up, whereas one scale per scene costs 17.6 cm: model size is not the bottleneck, scale is. Frame count governs coverage, not accuracy, and LiDAR confidence maps give no measurable gain.

TABLE IX
PER-FRAME SCALE OF THE PRETRAINED METRIC MODEL VS.
FT-LiDAR-24 (SIX TEST ROOMS, EVERY FIFTH FRAME).

	Pretrained	FT-LiDAR-24
median(GT/pred) per room	0.69–0.85	0.93–1.21
per-frame swing within a room (p95/p5)	1.55×	1.52×
AbsRel, raw	0.381	0.134
AbsRel, one (S, T) per room from 10 GT frames	0.127	0.117
AbsRel, right scale on every frame	0.045	0.051
AbsRel, oracle (s, t) per frame	0.035	0.042

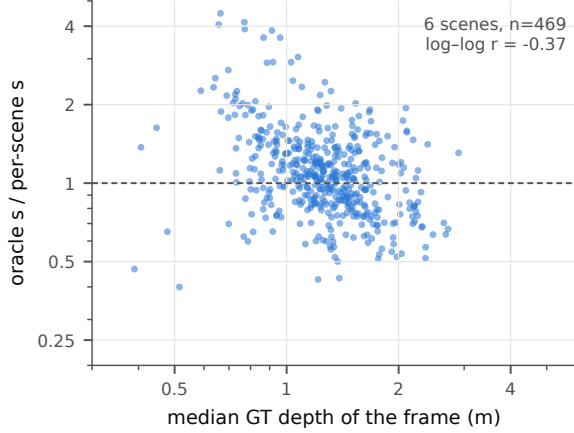


Fig. 4. Per-frame oracle scale divided by the per-scene scale (DA-v2 Large, all scenes) against the frame’s median GT depth: the scale a frame needs depends on how close the camera is to what it sees.

VI. DISCUSSION

A. Why fine-tuning works—and why it is not enough

We analysed every fifth frame of the six test rooms (471 frames): per frame, the GT/pred ratio and AbsRel under increasing amounts of scale help (Table IX). **Fine-tuning learns the domain’s multiplier.** The pretrained model is 15–31 % too far in every room; FT-LiDAR-24 is within $\sim 20\%$ in both directions. FT-LiDAR-24 without calibration (0.134) is on par with the pretrained model given a per-room scale fitted on ground truth (0.127): what a one-off calibration provides has moved into the head’s weights, learned from LiDAR in other rooms. With the encoder frozen, shape barely changes (oracle 0.035 vs. 0.042). **The per-frame swing remains.** Within one room, frames at p5 and p95 still need factors $\sim 1.5\times$ apart, as before fine-tuning; with the right scale on every frame FT-LiDAR-24’s AbsRel would drop to 0.051. That information is not in a single image and does not come from a better teacher; it must come from metric data at run time. Fine-tuning and per-frame metric points are therefore complementary.

B. Scale follows the view, not time

Refitting (s, t) on every fifth frame of the relative model shows why one calibration per scene fails. Within one scan the central 90 % of frames need scales 2.6–4.9 $\times$ apart; the correlation of scale with time flips sign between scenes (-0.33

TABLE X
USE-CASE FIT: RECALL@ τ OVER 6 SCENES ($\checkmark \geq 0.9$, ~ 0.75 – 0.9 , $\times < 0.75$).

Use case (τ)	LiDAR	sparse	per_scene	FT-LiDAR-24	metric
overview (10 cm)	$\checkmark$ 0.97	$\checkmark$ 0.91	$\times$ 0.47	$\times$ 0.51	$\times$ 0.10
furniture (5 cm)	$\checkmark$ 0.91	$\sim$ 0.77	$\times$ 0.27	$\times$ 0.28	$\times$ 0.04
renovation (2 cm)	$\times$ 0.52	$\times$ 0.39	$\times$ 0.09	$\times$ 0.09	$\times$ 0.01

to $+0.30$), so this is *not cumulative drift*, whereas the correlation with the frame’s median depth is negative in every scene (-0.27 to -0.75 , Fig. 4). The worst frames are downward shots of table-tops at 0.5–0.8 m and frame-filling blank walls or cabinets: with no size cue the model guesses at “room scale,” and a table at 0.5 m lands at 1.5–2 m as a phantom surface. The metric model behaves the same way: right on average up to a per-room bias, wrong per frame (GT/pred from 0.43 to 1.35 within one room), but with almost no shift term—one scalar per frame recovers 97 % of its gap to the oracle. In 3D the same surface seen from several frames is placed at several distances and TSDF averages them into a thick slab.

Rooms that are hard for LiDAR are not those that are hard for monocular depth: the standard bedroom 47333774 is the easiest for LiDAR (2.4) but second-hardest for mono+oracle (6.2), because a small room scanned close to blank walls gives the model no cue. The mirrored bathroom is hard for every source whose scale is right (LiDAR 4.1, oracle 7.5), so a bathroom-renovation product should expect worse than the averages.

C. Implications for a product

Table X translates the results into product tiers. iPad LiDAR suffices for furniture placement but not for renovation measurement. On a device without LiDAR, neither one-off calibration nor a raw metric model serves any tier; FT-LiDAR-24 raises recall@10 cm from 0.10 to 0.51 with nothing at run time, a large step that is still short of the overview tier. With per-frame sparse points the same device reaches 6.0 cm, close to furniture placement. The order of investment the data suggest is therefore: **(1) per-frame scale from VIO points, (2) a fine-tuned model to remove the domain bias, (3) model size last**—investing in the model first returns < 3 cm on a 22 cm base. Strategically, an app with Pro-device users can collect its own teacher during normal use (RGB, LiDAR depth and pose arrive together from ARKit), replacing a \$20k laser scanner; our data show no accuracy case for laser capture for this purpose. At 4 fps DA-v2 Large takes $3.6\times$ the video length on a laptop and DA-v2 Small $0.5\times$; fusion is $< 3\%$ of the time.

VII. LIMITATIONS AND FUTURE WORK

Reference and poses. The reference mesh is fused from Faro depth with the same poses as every row, so pose drift is excluded and a real phone system incurs additional error; completeness means complete relative to what Faro saw. **Scale**

of the evaluation. Mesh numbers come from six rooms (one used in development) and 2D numbers from 26, all captured with one iPad model in homes; offices, empty rooms, low light and other phone cameras are absent. **Fine-tuning.** Checkpoints were selected on only three validation rooms, whose ranking (FT-LiDAR-24 > FT-LiDAR-10 > FT-Faro-10) does not reproduce on the test rooms; FT-Faro-10 and FT-LiDAR-10 differ in the number of views as well as the label; and only the head of one model was trained with one recipe. **Proxies and bounds.** `per_scene` calibrates on ground truth and `sparse_points` uses clean LiDAR samples instead of triangulated VIO features, so both are upper bounds. Depth Pro was run through one implementation on 640×480 input. Times are unoptimised PyTorch on Apple Silicon and are meaningful only as ratios.

Future work, in order of impact: (1) combine FT-LiDAR-24 with per-frame sparse points, which the 2D analysis bounds at AbsRel 0.051; (2) a controlled teacher comparison on identical frames and a larger validation set; (3) training data and evaluation on more phone cameras, including Android; (4) evaluation with real ARKit/ARCore feature points and a reference captured by our own app.

VIII. CONCLUSION

We asked whether fine-tuning a metric monocular depth model with a phone's own LiDAR makes it better than its pretrained self, measured against a laser scanner, through a room-reconstruction pipeline in which the depth source is the single variable. **It does, in every room we can measure:** fine-tuning only the head of DA-v2 Metric-Indoor on 24 rooms (FT-LiDAR-24) lowers Chamfer from 54.3 cm to 15.7 cm on all six held-out rooms and per-frame AbsRel from 0.410 to 0.130 on 20 further rooms, with no calibration or sensor at run time. **The teacher need not be a laser scanner:** Faro and LiDAR teachers are not measurably different, and knowing the camera's intrinsics is no substitute for in-domain labels. **It is not yet a sensor:** iPad LiDAR gives 3.0 cm and a monocular model scaled per frame on 200 sparse points 6.0 cm; FT-LiDAR-24's remaining error is a scale that swings with what each frame shows, which fine-tuning de-biases but cannot remove. The path to a usable product on devices without LiDAR is therefore a fine-tuned model combined with per-frame metric points from the device's tracker.

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